

DISCUSSION

ORIGIN OF THE ROAN ANTELOPE COPPER DEPOSIT OF NORTHERN RHODESIA

Sir: In the September–October issue (pp. 575–616) of *ECONOMIC GEOLOGY* there is a paper on the Roan Antelope Mine and reference is made to an earlier article by Garlick and Brummer. There have also appeared articles and discussions on the genesis of the Northern Rhodesian copper ores in the *Bulletin of the Institution of Mining and Metallurgy*. The authors have advanced arguments for a sedimentary origin of the copper ores. As I understand the argument it is as follows: The ancient schists were intruded by granite and mineralization accompanied the closing phases of the granite intrusion. Weathering and erosion followed and the sediments of the Mine Series were laid down in depressions adjacent to the areas being eroded. The copper minerals were carried down and deposited at the same time partly as a result of biochemical processes.

Folding and metamorphism followed this deposition and the last event was igneous activity in the form of gabbro intrusions.

The authors repudiate the earlier contention that there was a granite which cut the Mine Series. I was on the copper belt for part of 1929 to 1931 and saw most of the deposits in Northern Rhodesia and the Haut Katanga. I had the pleasure of driving Dr. Schneiderhöhn from N'Dola to N'Kana and on to N'Changa, so I heard a good deal about syngenetic ores from him. My own views at that time were definitely on the side of those who postulated an epigenetic origin, and there were many lively discussions with Bancroft, Gray, Davidson, Bateman and Grout, who all held that the ores were epigenetic. I should like to point out the difficulties that those who advocate the sedimentary hypothesis will have to explain.

First—Bancroft always stressed the importance of drag folds as being associated with the ore zones. This association seemed to be remarkably exhibited at N'Kana, N'Changa, Chambishi, and at the Roan Antelope. If these ores are epigenetic they are found in those places exactly where they should be.

For those who favor sedimentation it will be necessary to explain this strange fact, that the drag folds occurred where the ore had previously been deposited.

Secondly—There is no possible way to explain the Kipushi Mine where the ore occurs between the faulted ends of the *Série des Mines* and the fault plane, by an appeal to washed in sediments. I have still some of that remarkable ore in our museum.

The character of the ore in the River lode at N'Changa in my opinion is clearly epigenetic where veinlets of copper minerals run up along the shales

and cross from one shale bed to another. The recent authors attempt to explain this jumping of the beds by an appeal to metamorphism. It is a difficult task.

Thirdly—From polished sections and from various places underground which I saw, some twenty years ago, replacement was apparent.

Fourthly—The Mine Series consists of sediments that range from conglomerates and arkoses at the base to shales and dolomitic shales at upper horizons. The copper minerals are found in the shales. If the cupreous minerals are derived from a previous mineralization by erosion and transport, it seems very curious that these minerals are not with the coarse basal sediments but are associated with the mudstones on a higher horizon. The grain sizes given by Graton in a report in possession of the Roan Antelope Mine are as follows:

Relatively few grains	0.60 mm.
Average grain	0.06 mm.
Very many as small as	0.006 mm.
Many as small as	0.0025 mm.

If the average grain diameter of chalcocite is 0.06 mm. the average grain of the detrital quartz associated with it should be 0.08 mm. In other words the grain size of the sediments should be larger than that of the copper minerals which is not the case. The average grain size of sediments of the type of the Roan Shales is 0.01 mm.

If the chalcocite or chalcopyrite is the result of sedimentary processes these minerals should have been found in the conglomerates or arkoses at the base of the Mine Series. Those who favor a sedimentary origin have therefore to explain why they are in the mudstones well above the basal beds.

Fifthly—The statement that *all* the granite is older than the Mine Series is open to dispute. In my opinion the N'Changa granite is younger. The evidence is two-fold: At the nose of the syncline there is a tongue of granite or pegmatite cutting into the beds.

In the footwall rocks of the long crosscut to the first shaft which was put down in 1930 or 1931 the feldspathic sandstones as I remember them show progressive feldspathization, i.e., granitization as the granite is approached.

Far be it from me to discourage ideas—but hypotheses are tools not creeds, and those who advance ideas must be prepared to defend them.

My testament regarding the copper belt is therefore as follows: The ores are hypogene and epigenetic and were introduced into the sediments after those rocks had been mountain built and intruded by igneous rocks.

Tot homines quot sententiae.

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